

Emergency Stop Category 1 using Safe Stop 1 (SS1) and Safe Brake Control (SBC) via GuardLogix and PowerFlex 755 Drive

Products: GuardLogix 5580, PowerFlex 755-S4 Drive, Rotary Induction Motor-Encoder Feedback

Safety Rating: Cat. 3, PLd to ISO 13849-1: 2023



Topic	Page
Important User Information	2
General Safety Information	3
Introduction	3
Use Sample Project Files	5
Safety Function Realization: Risk Assessment	5
Emergency Category 1 Stop Safety Function	5
Safety Function Requirements	6
Bill of Materials	7
Setup and Wiring	7
Configuration	9
Programming	14
Calculation of the Performance Level	25
Verification and Validation Plan	27
Additional Resources	27

Important User Information

Read this document and the documents listed in the additional resources section about installation, configuration, and operation of this equipment before you install, configure, operate, or maintain this product. Users are required to familiarize themselves with installation and wiring instructions in addition to requirements of all applicable codes, laws, and standards.

Activities including installation, adjustments, putting into service, use, assembly, disassembly, and maintenance are required to be carried out by suitably trained personnel in accordance with applicable code of practice.

If this equipment is used in a manner not specified by the manufacturer, the protection provided by the equipment may be impaired.

In no event will Rockwell Automation, Inc. be responsible or liable for indirect or consequential damages resulting from the use or application of this equipment.

The examples and diagrams in this manual are included solely for illustrative purposes. Because of the many variables and requirements associated with any particular installation, Rockwell Automation, Inc. cannot assume responsibility or liability for actual use based on the examples and diagrams.

No patent liability is assumed by Rockwell Automation, Inc. with respect to use of information, circuits, equipment, or software described in this manual.

Reproduction of the contents of this manual, in whole or in part, without written permission of Rockwell Automation, Inc., is prohibited.

Throughout this manual, when necessary, we use notes to make you aware of safety considerations.



WARNING: Identifies information about practices or circumstances that can cause an explosion in a hazardous environment, which may lead to personal injury or death, property damage, or economic loss.



ATTENTION: Identifies information about practices or circumstances that can lead to personal injury or death, property damage, or economic loss. Attentions help you identify a hazard, avoid a hazard, and recognize the consequence.

IMPORTANT Identifies information that is critical for successful application and understanding of the product.

Labels may also be on or inside the equipment to provide specific precautions.



SHOCK HAZARD: Labels may be on or inside the equipment, for example, a drive or motor, to alert people that dangerous voltage may be present.



BURN HAZARD: Labels may be on or inside the equipment, for example, a drive or motor, to alert people that surfaces may reach dangerous temperatures.



ARC FLASH HAZARD: Labels may be on or inside the equipment, for example, a motor control center, to alert people to potential Arc Flash. Arc Flash will cause severe injury or death. Wear proper Personal Protective Equipment (PPE). Follow ALL Regulatory requirements for safe work practices and for Personal Protective Equipment (PPE).

General Safety Information

Contact Rockwell Automation to learn more about our safety risk assessment services.

IMPORTANT

This application example is for advanced users and assumes that you are trained and experienced in safety system requirements.



ATTENTION: Perform a risk assessment to make sure that all task and hazard combinations have been identified and addressed. The risk assessment can require additional circuitry to help reduce the risk to a tolerable level. Safety circuits must consider safety distance calculations, which are not part of the scope of this document.

Safety Distance Calculations



ATTENTION: While safety distance or access time calculations are beyond the scope of this document, compliant safety circuits must often consider a safety distance or access time calculation.

Non-separating safeguards provide no physical barrier to help prevent access to a hazard. Publications that offer guidance for calculating compliant safety distances for safety systems that use non-separating safeguards, such as light curtains, scanners, two-hand controls, or safety mats, include the following:

- EN ISO 13855:2010 (Safety of Machinery – Positioning of safeguards with respect to the approach speeds of parts of the human body)
- EN ISO 13857:2019 (Safety of Machinery – Safety distances to help prevent hazardous zones being reached by upper and lower limbs)
- ANSI B11:19 2019 (Machines – Performance Criteria for Safeguarding)

Separating safeguards monitor a movable, physical barrier that guards access to a hazard. Publications that offer guidance for calculating compliant access times for safety systems that use separating safeguards, such as gates with limit switches or interlocks (including SensaGuard™ switches), include the following:

- EN ISO 14119:2013 (Safety of Machinery – Interlocking devices associated with guards – Principles for design and selection)
- EN ISO 13855:2010 (Safety of Machinery – Positioning of safeguards with respect to the approach speeds of parts of the human body)
- EN ISO 13857:2019 (Safety of Machinery – Safety distances to help prevent hazardous zones being reached by upper and lower limbs)
- ANSI B11:19 2019 (Machines – Performance Criteria for Safeguarding)

In addition, consult relevant national or local safety standards to verify compliance.

Introduction

This document explains how to wire, configure, and program a GuardLogix® controller to interface with a PowerFlex® 755 integrated safety function option module (20-750-S4) to perform monitored SS1 (Safe Stop 1) and SBC (Safe Brake Control) functions. The safety function option module works with the PowerFlex 755 Dual Encoder Feedback module (catalog number 20-750-DENC) for example, to transfer the encoder feedback position into the Safety Task. The GuardLogix Safety Task contains logic to monitor the encoder feedback position (SFX), perform a monitored Safe Stop 1 (SS1), perform a Safe Brake Control function (SBC) and initiate a Safe Torque Off request (STO) when requested. This logic in the Safety Task is initiated when the E-stop input is pressed, and the E-stop condition is active.

The actuator is a rotary induction motor that the PowerFlex 755 module controls. Any malfunction that is detected by the SS1 or SBC function initiates a STO request that is followed with an SBC brake release request.

This example uses a GuardLogix controller, but you can substitute a Compact GuardLogix controller that supports the safety rating that is demonstrated in this safety function application technique. The Safety Integrity Software Tool for the Evaluation of Machine Applications (SISTEMA) calculations that are shown later in this document must be recalculated if different products are used. This example uses a catalog number 20-750-DENC-1 encoder interface card. A catalog number 20-750-UFB-1 universal feedback option module can also be used.

Application of a Safety Brake

A few characteristics of a safety brake that are relevant to this application technique include the following:

- When an application requires a means to engage (grab) a load so it can be safely held, a safety brake can be used on the actuator.
- The safety brake is designed to stop and hold a motor that is attached to a load.
- Each safety brake manufacturer provides guidance for the maintenance of their safety brake. They also provide the maximum number of actuation cycles between inspections. When a safety brake stops a moving load, it counts against its maximum number of actuation cycles, after which the safety brake needs inspection for replacement or maintenance. If the load is stationary (zero speed) when the safety brake is engaged, this actuation instance does not count against the replacement or maintenance criteria.

Application Example Considerations

For all examples, deliberate consideration was made for the application of the safety brake when the motor is being enabled (restarting) and when the motor is being disabled. If the SBC safety instruction was solely used to stop and disable the motor, the maximum actuation limit would likely be achieved quickly and maintenance would become inefficient as the SBC alone does not consider standstill speed.

Additionally, we consider the normal (typical) safety brake use, that is, we use the SS1 (Safe Stop 1) instruction and standard logic to decelerate the motor to a configurable standstill speed, then engage the safety brake with the SBC instruction. These considerations are used to prolong the actuation cycle limit while safely engaging the load.

Mechanical Fastening of the Encoder

A physical connection between the motor and encoder, and between motor and load is implied in this publication.



ATTENTION: While mechanical fastening of the encoder is beyond the scope of this publication, the physical connection between the motor and encoder, and between the motor and load and/or between the encoder and the load, must be evaluated in a mechanical safety assessment.

The detachment of a fastening method, which results in misrepresentation of the electrically derived position versus a mechanical position that could potentially lead to a dangerous failure, must be evaluated. Publications that offer guidance for mechanical fastening include the following:

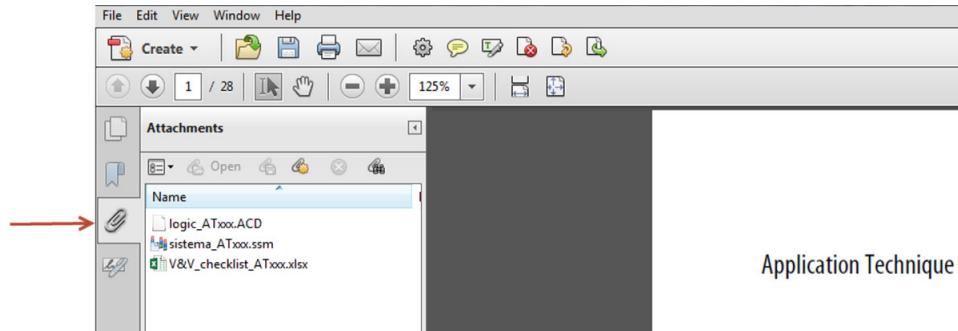
- IEC 61800-5-3 (Adjustable speed electrical power drive systems - Safety requirements for encoders - Functional, Electrical, and Environmental)
- GS-IFA-M21 (IFA white paper) - (Principles for the testing and certification of rotary and position-measuring systems for functional safety)
- Annex A of EN ISO 13849-2 - (Validation tools for mechanical systems)

Use Sample Project Files

Sample project files (ACD, SISTEMA, and Verification and Validation checklist) are attached to this document to help you implement this safety function.

To access these files, follow these steps.

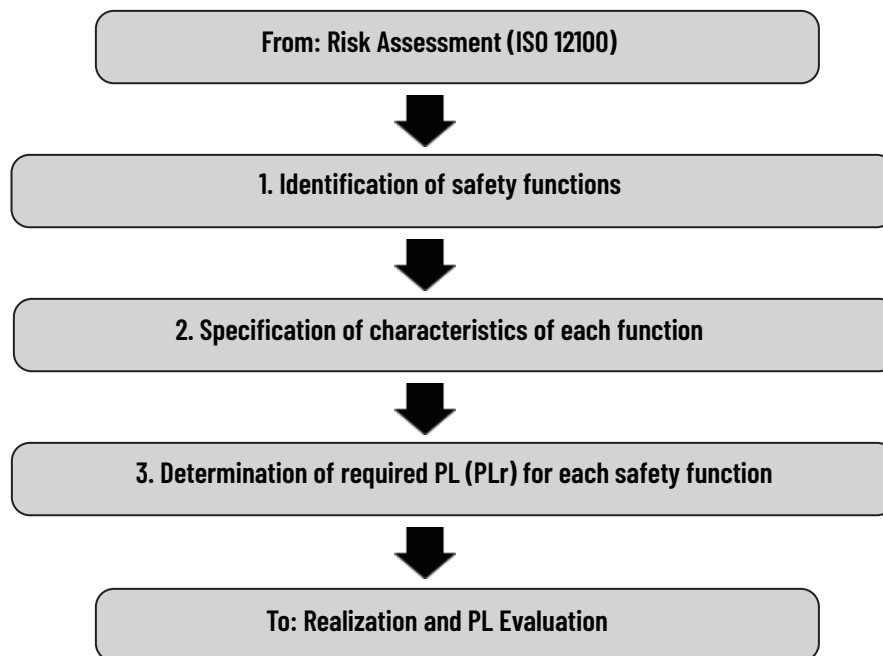
1. If you are viewing the PDF file in a browser and do not see the Attachments link , download the PDF file and open it in the Adobe Acrobat Reader application.
2. Click the Attachments link .
3. Right-click and save the desired file.



4. Open the file in the appropriate application.

Safety Function Realization: Risk Assessment

The Performance Level required (PLr) is the result of a risk assessment and refers to the amount of the risk reduction to be carried out by the safety-related parts of the control system. Part of the risk reduction process is to determine the safety functions of the machine. In this application, the Performance Level required by the risk assessment is category 3, Performance Level d (cat. 3, PLd), for each safety function. A safety system that achieves cat. 3, PLd, or higher, can be considered control reliable. Each safety product has its own rating and can be combined to create a safety function that meets or exceeds the PLr.



Emergency Category 1 Stop Safety Function

This application technique includes one safety function:

1. Emergency Stop Function: Safe Stop 1 (SS1) faults to Safe Brake Control (SBC)

Safety Function Requirements

The emergency stop (E-stop) function is complementary to other safeguards on the machine.

Emergency Stop Function: Safe Stop 1 (SS1) followed by Safe Brake Control (SBC)

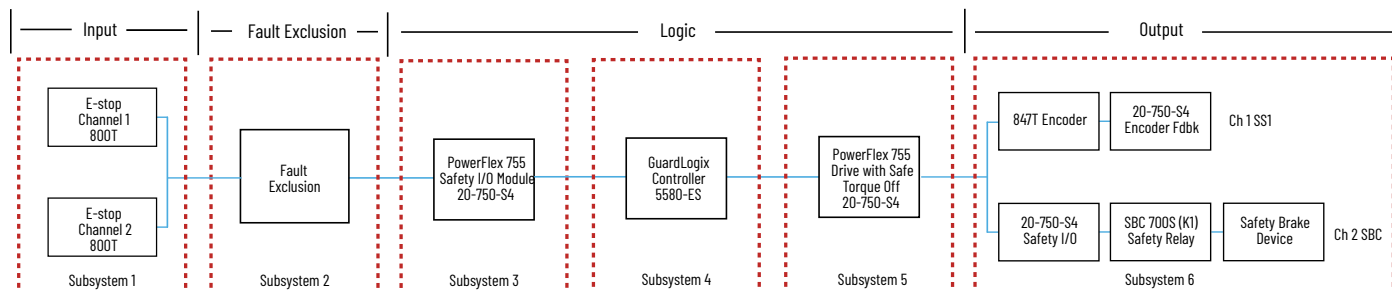
In normal operation, the E-stop request (Input) is used to request a Safe Stop 1 (SS1) that monitors deceleration until the motor reaches the configured standstill speed. At standstill speed, the Safe Brake Control (SBC) request occurs. The SBC logic de-energizes a safety relay that engages the Safety Brake device. Next, a Safe Torque Off (STO) request initiates from the SBC instruction. When the STO request is complete and active (Module01:S0.STOOutput=OFF), the motor is not capable of torque production.

Emergency Stop Function: Safe Stop 1 (SS1) faults to Safe Brake Control (SBC)

When the E-stop request (Input) is used to initiate a safe stop and:

1. The safety logic detects a fault with the SS1 operation, the safety logic requests a Drive STO (Safe Torque Off; SS1.STO_Req) to remove the torque producing capability from the motor. Simultaneously, the safety logic engages the Safety Brake device (SBC.ReleasePerm). The safety brake device is engaged to stop motor rotation.
2. The safety logic detects a fault with the SBC operation, the safety logic requests the STO (Safe Torque Off, SBC.STO_Req) to remove the torque producing capability from the motor. The safety logic engages the Safety Brake device (SBC.ReleasePerm) and the safety brake device is engaged to stop motor rotation.

Hazardous motion cannot be restarted until the E-stop is released and the Reset button is pressed and released. If the SBC or SS1 instructions fault, recovery requires two resets. After clearing the SBC and/or SS1 faulting conditions, the first reset clears instruction faults. After releasing the E-stop, the second reset enables the run permissive.



The safety functions in this application technique each meet or exceed the requirements for category 3, Performance Level d (cat. 3, PLd), per ISO 13849-1 and control reliable operation per ANSI B11.19.

Considerations for Safety Distance and Stopping Performance

Based on the selection of a sensor subsystem, the risk assessment determines if a safety distance calculation is required. Typically, a safety distance calculation is required if a non-separating sensor subsystem (such as a light curtain) or guard doors without a safety-rated locking function is part of the safety function. If a safety distance calculation is required for this safety function, the following documents can be referenced:

- GuardLogix 5580 and Compact GuardLogix 5380 Controller Systems Safety Reference Manual, publication [1756-RM012](#)
- Machinery SafeBook 5 – Safety-related control systems for machinery, publication [SAFEBC-RM002](#)
- Light Curtain and GuardLogix Controller Safety Function Application Technique, publication [SAFETY-AT191](#)

Bill of Materials

This application technique uses these products.

Quantity	Cat. No.	Description
1	20G-xxx	PowerFlex 755 drive
1	20-750-DENC-1	PowerFlex 750 dual incremental feedback module
1	20-750-S4	PowerFlex 750 integrated safety functions option module
1	—	AC rotary induction motor (application dependent)
1	—	Safety brake (application dependent)
1	847T-DL1A-RP02048	Incremental encoder (application dependent); specs: 4.5...5.5V DC differential line driver, TTL (B-leads-A, CW, Z gated), minimum 2048 pulses/revolution
1	800T-FXM6A7	30.5 mm (1.20 in.) Type 4/13 3-position push button, non-illuminated, mushroom head (push-pull), red, OUT/CNTR/INMom/Maint/Maint, 1 N.C.L.B.-1 N.C. (E-stop button)
1	700S-CF620QJBC	Safety control relay, 8-pole, 3 N.O./1 N.C. base, 1 N.O./3 N.C. auxiliary, bifurcated contact, 24V DC (with electric coil)
1	1606-XLP72E	Compact power supply, 24...28V DC, 72 W, 120/240V AC 1

Choose one of the following safety-controller hardware groups.

Quantity	Controller	Cat. No.	Description
1	GuardLogix 5580	1756-L81ES	GuardLogix Processor, 3 MB standard memory, 1.5 MB safety memory
		1756-L82ES	GuardLogix Processor, 5 MB standard memory, 2.5 MB safety memory
		1756-L83ES	GuardLogix Processor, 10 MB standard memory, 5 MB safety memory
		1756-L84ES	GuardLogix Processor, 20 MB standard memory, 6 MB safety memory
1		1756-PA72	Power supply, 120/240V AC input, 3.5 A @ 24V DC
1		1756-A7	Seven-slot ControlLogix® chassis
1	Compact GuardLogix 5380-SIL 2	5069-L306ERS2	Compact GuardLogix processor, 0.6 MB standard memory, 0.3 MB safety memory
		5069-L306ERMS2	
		5069-L310ERMS	Compact GuardLogix processor, 1.0 MB standard memory, 0.5 MB safety memory
		5069-L310ERMS2	
		5069-L320ERS2	Compact GuardLogix processor, 2.0 MB standard memory, 1.0 MB safety memory
		5069-L320ERMS2	
		5069-L330ERS2	Compact GuardLogix processor, 3.0 MB standard memory, 1.5 MB safety memory
		5069-L330ERMS2	
		5069-L340ERS2	Compact GuardLogix processor, 4.0 MB standard memory, 2.0 MB safety memory
		5069-L340ERMS2	
		5069-L350ERS2	Compact GuardLogix processor, 5.0 MB standard memory, 2.5 MB safety memory
		5069-L350ERMS2	
		5069-L380ERS2	Compact GuardLogix processor, 8.0 MB standard memory, 4.0 MB safety memory
		5069-L380ERMS2	
		5069-L3100ERS2	Compact GuardLogix processor, 10.0 MB standard memory, 5.0 MB safety memory
		5069-L3100ERMS2	
1		5069-RTB64-SCREW	5069 Compact I/O™ power terminal RTB kit for both 4- and 6-pin screw

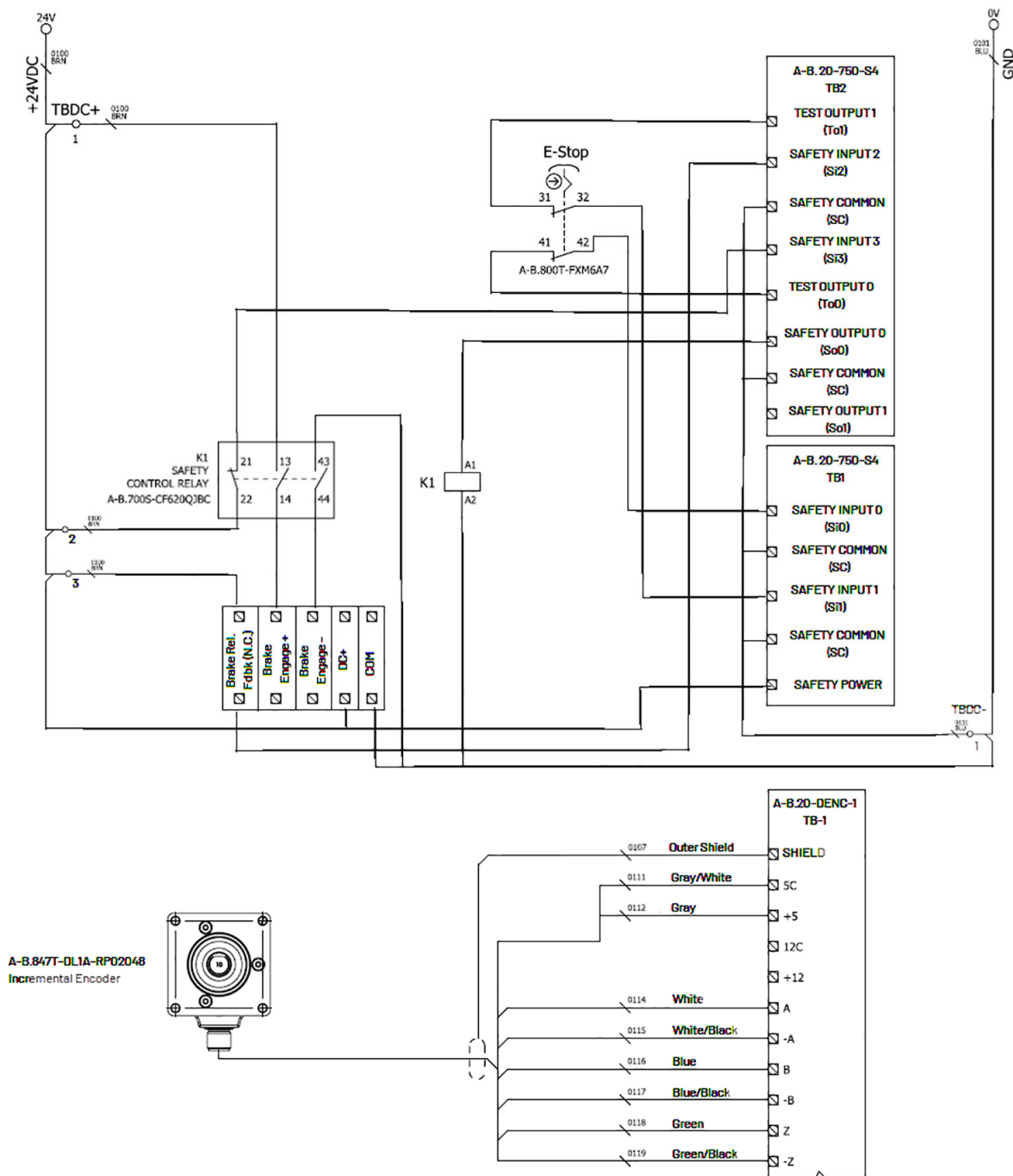
Setup and Wiring

For detailed information on how to install and wire the products in this application technique, refer to the publications that are listed in the [Additional Resources on page 27](#).

System Overview

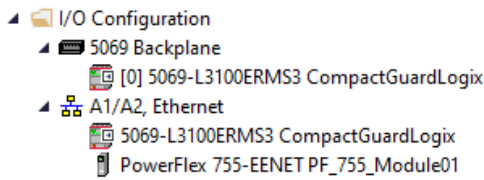
In this document, the PowerFlex 755 integrated safety function option module (catalog number 20-750-S4) is used for all Safety Input and Output wiring. The E-stop button is wired to the PowerFlex 755 option module, Safety Inputs 0 and 1. These inputs are monitored using the safety logic system and when the E-stop button is pressed, the safety logic initiates an SS1 (Safe Stop 1) instruction with monitored deceleration. When the SS1 is initiated, the standard logic decelerates the motor to zero speed. The Deceleration ramp is monitored with the SS1 instruction.

Rockwell Automation Publication SAFETY-AT203A-EN-P - April 2024

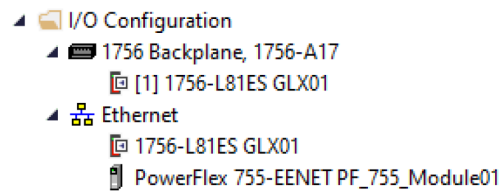


Network Architecture

GuardLogix 5380 Controller with local Guard I/O™



GuardLogix 5580 Controller with remote Guard I/O ⁽¹⁾



(1) When using a GuardLogix 5580 controller, slot 2 is reserved for the safety partner, which is required for SIL 3/PLc applications.

Configuration

The GuardLogix 5580ES controller is configured with the Studio 5000 Logix Designer® application (version 35 or later). You must create a project and add the PowerFlex 755 drive and peripherals. A detailed description of each step is beyond the scope of this document. Knowledge of the Logix Designer application is assumed.

For a Studio 5000 Logix Designer project file that you can import into your own application, see the attached ACD file. The attached ACD file includes a GuardLogix 5580 controller, but if you choose a 5380 controller, you can change the controller in the Logix Designer application. For instructions on how to access the attachments, see [Use Sample Project Files on page 5](#).

Minimum Software Version	Product
Logix Designer application version 35	GuardLogix 5580 or Compact GuardLogix 5380 controller
Firmware revision 1.xx	PowerFlex 755 20-750-S4 Integrated Safety Option Module
Add-on Profile version 16.002	PowerFlex 755 240V (4.2 ND) EENET

IMPORTANT Only the safety-related programming and configuration are discussed in this document. Standard motion control required to satisfy the safety monitoring functions are out of scope for this document.

Create a Project with a GuardLogix Controller

If you are not using the attached ACD file, follow these steps to create a project. For instructions on how to access the attachments, see [Use Sample Project Files on page 5](#).

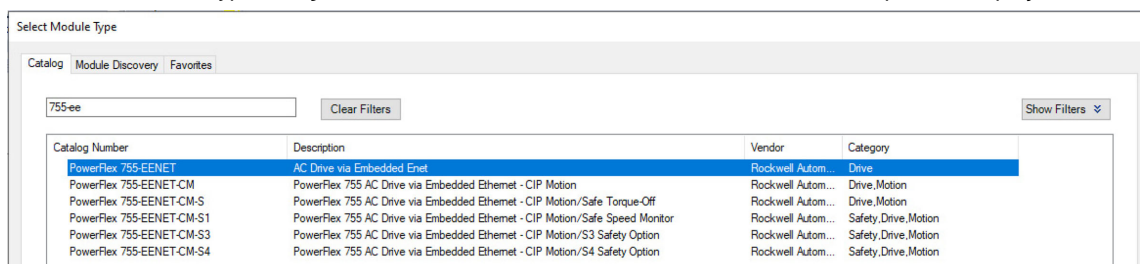
1. In the Logix Designer application, create a project with a GuardLogix controller.

IMPORTANT If you use a GuardLogix 5580 controller, you must configure the safety level of the controller on the Safety tab of the Module Properties dialog box. The default setting is SIL 2, PLd. For SIL 3, PLc operation, you must have a 1756-L8SP Safety Partner installed to the right of the primary controller.

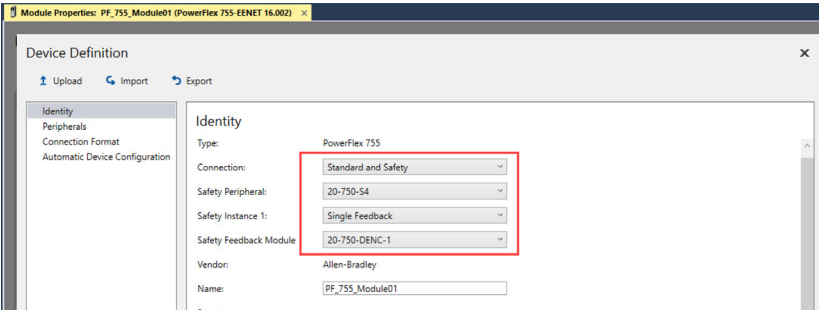
2. Set the IP address for the controller or any Ethernet communication modules, if used.
See [Additional Resources on page 27](#) for information on your I/O modules.

PowerFlex 755 Drive - Device Definition

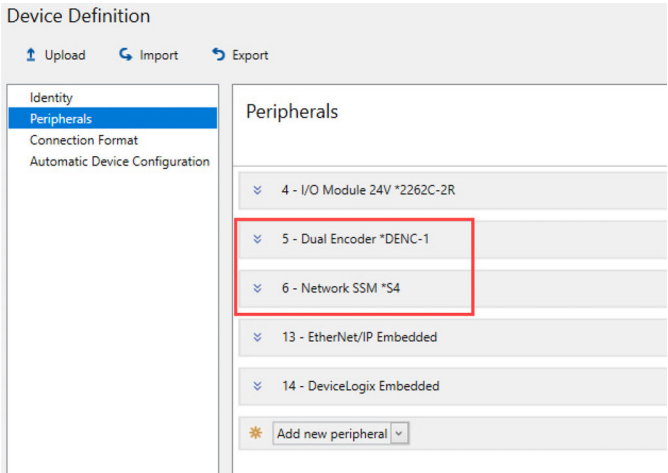
1. In the Select Module Type dialog box, add a 755-EENET (Standard I/O mode, not CIP™) adapter to the project I/O configuration.



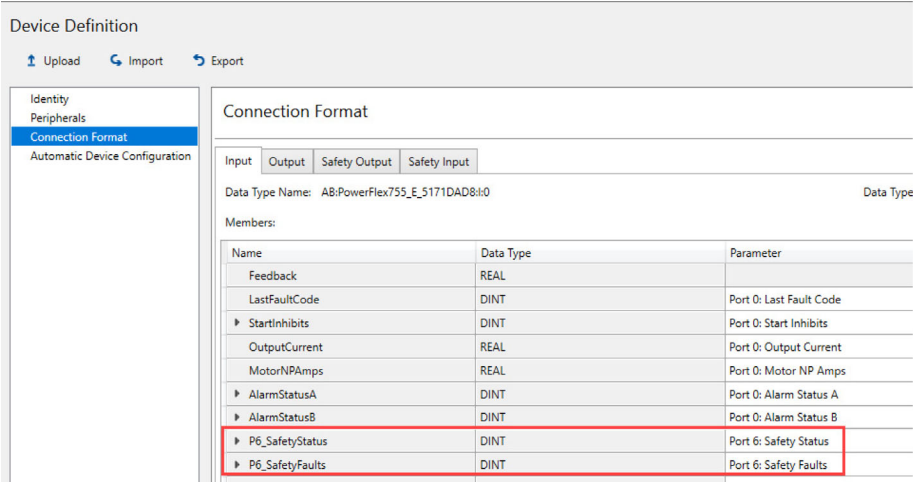
2. In the Module Properties dialog box, Device Definition screen, Identity panel, configure the Connection, Safety Peripheral, Safety Instance 1, and Safety Feedback Module fields as shown in the graphic.



3. Click Peripherals and verify that the safety-related peripherals are configured in the appropriate slot locations.

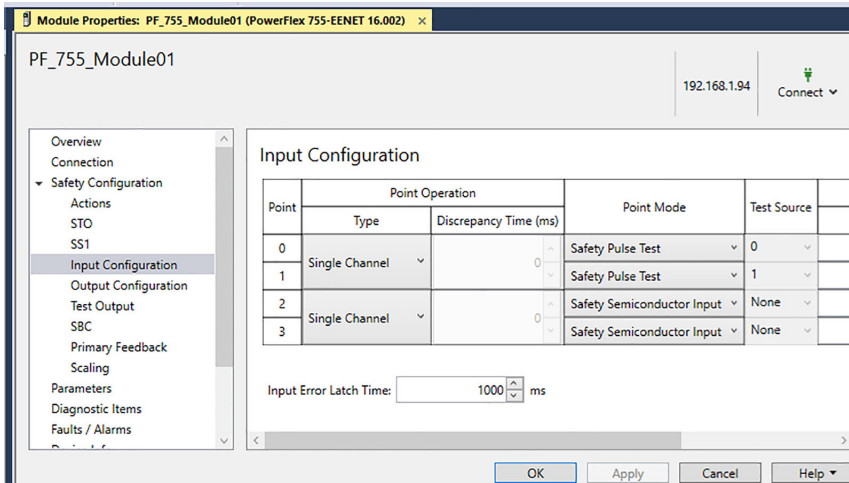


4. Click Connection Format and add the SafetyStatus and SafetyFaults input datalinks.

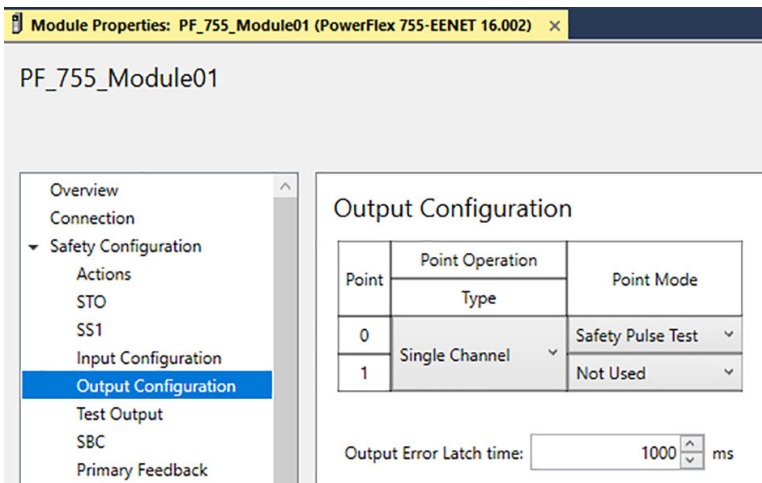


PowerFlex 755 - Initial Safety Configuration

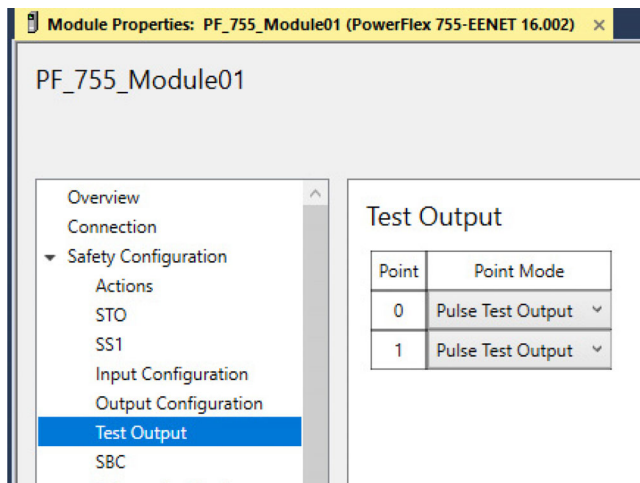
1. In the Module Properties dialog box do the following:
 - a. Expand Safety Configuration.
 - b. Click Input Configuration.
 - c. In the Point Mode column, configure the four points as shown.



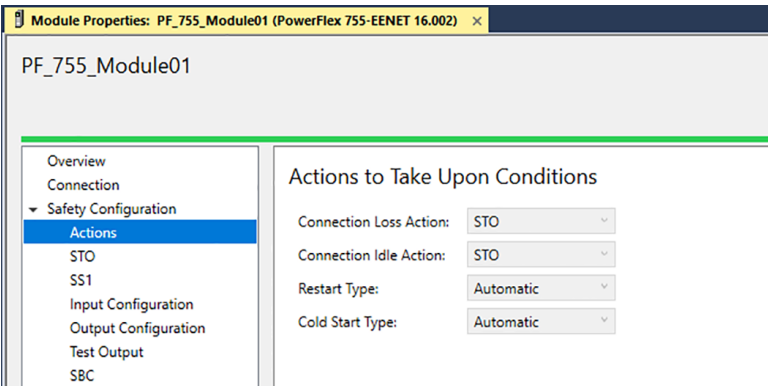
2. Click Output Configuration, and do the following:
 - a. Set the Type column, select Single Channel.
 - b. In the Point Mode column, select Safety for Point 0. This point is the only Output that is used for this example. If you require another configuration for Point 1, change it here.



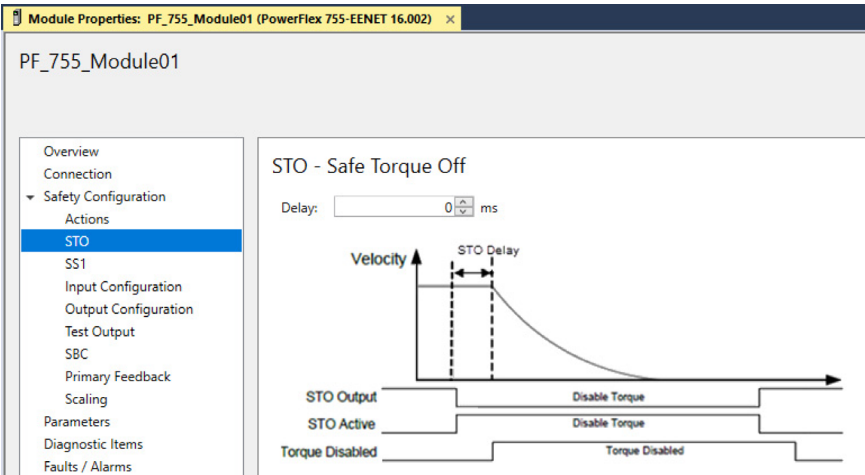
3. Click Test Output, and in the Point Mode column, select Pulse Test Output for Point 0 and Point 1.



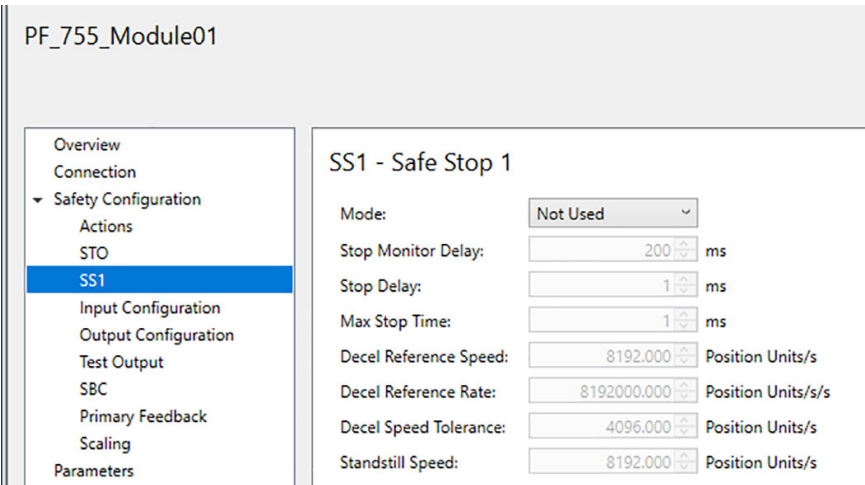
4. Click Actions. Set the Actions to take Upon Conditions as shown.



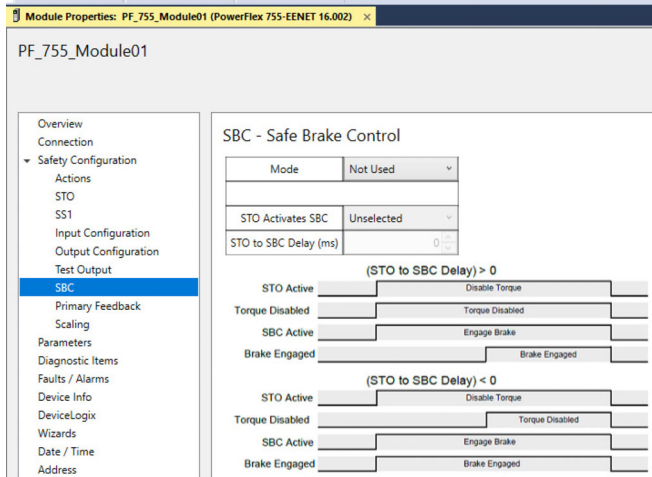
5. Click STO. Choose the Delay, if used. This delay represents the delay from when the STO request is acknowledged in the drive to the actual torque being disabled.



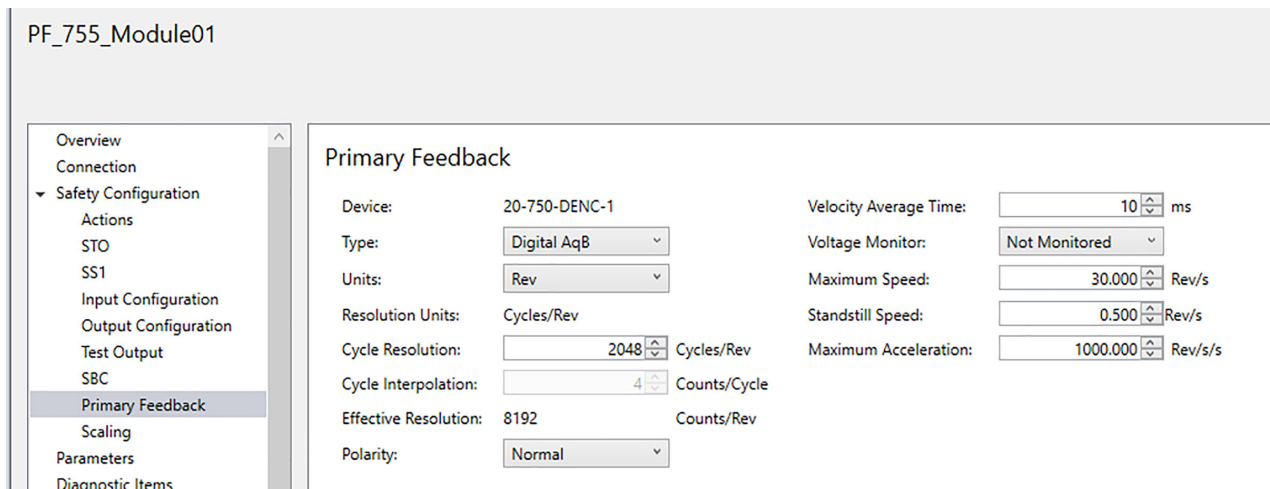
6. Click SS1. This example uses a controller-based SS1 safety instruction. The drive-based SS1 Mode is configured as Not Used.



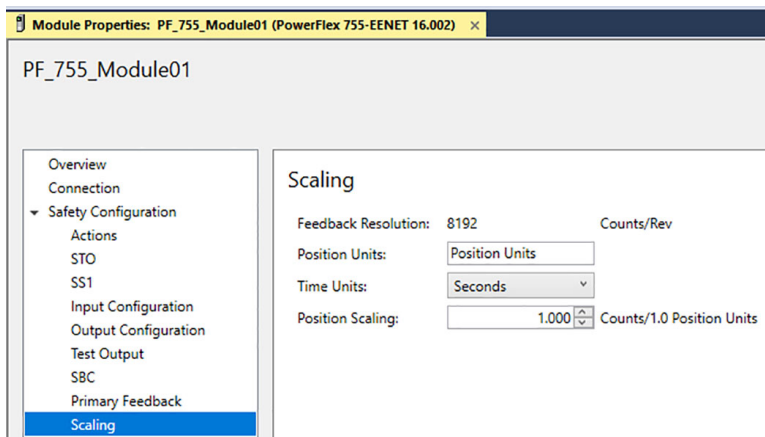
7. Click SBC and verify that the Mode is Not Used. In this example, the SBC request originates from the controller safety logic. The SBC configuration that is shown in the Add-on Profile of the drive is used when the drive is controlling the safety-braking device without using the controller.



8. Click the Primary Feedback and configure your encoder feedback as your application requires. In this example, we use 2048 Cycles/Rev which is an effective resolution of 8192 counts/rev.



9. Click Scaling, and do the following:
- In the Position Units field, choose your units. In this example, we use the default Position Units.
 - In the Position Scaling field, choose the scaling that you use. In this case, 1 count/1 Position Unit.



10. Click Parameters and set Port 5, Parameter 2 - Enc 0 PPR to the Feedback Resolution you configured in the previous step. In this example, the value for Parameter 2 is 2048 PPR.

PF_755_Module01

Overview
Connection
Safety Configuration
Parameters
Diagnostic Items
Faults / Alarms
Device Info
DeviceLogix
Wizards
Date / Time
Address

Parameters

5 - Dual Encoder All Parameters Show Non-Defaults Filter Value Reset Defaults Print Export

Port	#	Name	Value	Units	Internal Value	Default	Min	Max
5	1	Enc 0 Cfg	00000000 00000100		4	00000000 00000100		
5	2	Enc 0 PPR	2048	PPR	2048	1024	2	
5	3	Enc 0 FB Lss Cfg	FitCoastStop		3	FitCoastStop	0	
5	4	Enc 0 FB		0	0		-21474...	
5	5	Enc 0 Sts	00000000 00000000 00000000 00000000		0	00000000 00000000...		
5	6	Enc 0 Error Sts	00000000 00000000 00000000 00000000		0	00000000 00000000...		
5	7	Enc 0 PhsLss Cnt	0		0	0	0	0

11. In the same Parameters dropdown list, do the following:
- Select Network SSM.
 - Set Port 6, Host Parameter 13 - SS1/SS2 Actn Src to Controller

Module Properties: PF_755_Module01 (PowerFlex 755-EENET 16.002) x

PF_755_Module01

Overview
Connection
Safety Configuration
Actions
STO
SS1
Input Configuration
Output Configuration
Test Output
SBC
Primary Feedback
Scaling
Parameters
Diagnostic Items
Faults / Alarms
Device Info
DeviceLogix
Wizards
Date / Time
Address

Parameters

6 - Network SSM All Parameters Show Non-Defaults Filter Value Reset Defaults Print Export

Port	#	Parameter Ty	Name	Value	Units	Internal Value	Default
6	32	Device	Enc2 Accel	0		0	0
6	33	Device	Enc2 Fault Type	No Fault		1	No Fault
6	34	Device	Enc2 Unit	Revolution		0	Revolution
6	3	Host	Safety State	Testing		1	Testing
6	4	Host	Safety Status	00000000 00000000 00000000 00000000		0	00000000 00000000...
6	5	Host	Safety Faults	00000000 00000000 00000000 00000000		0	00000000 00000000...
6	6	Host	Safe Status Mfg	00000000 00000000 00000000 00000000		0	00000000 00000000...
6	7	Host	Safe Faults Mfg	00000000 00000000 00000000 00000000		0	00000000 00000000...
6	8	Host	Safety Data A	0		0	0
6	9	Host	Safety Data B	0		0	0
6	11	Host	STO Actn Src	Drive		0	Drive
6	12	Host	STO Stp Actn	Coast		0	Coast
6	13	Host	SS1/SS2 Actn Src	Controller		1	Drive
6	14	Host	SS1 Stp Actn	Ramp		1	Ramp

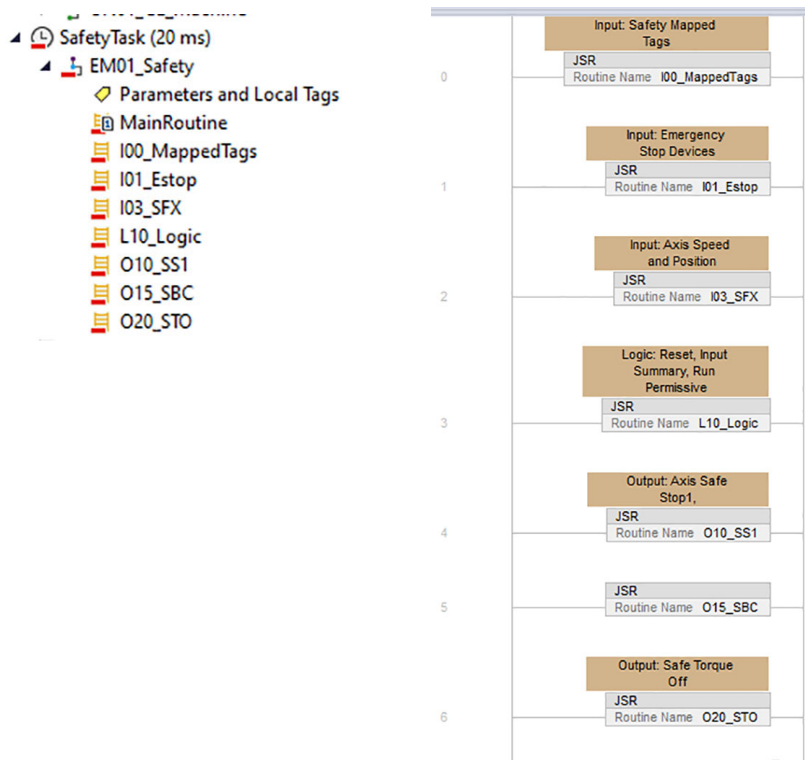
Programming

For controller logic that you can download to your controller, see the attached ACD file. For instructions on how to access the attachments, see [Use Sample Project Files on page 5](#).

Programming Safety

General Task Configuration

- For modularity, and following application software guidance from safety standards, the safety zone program has been broken into routines for input, logic, and output. Be sure to call all routines from the MainRoutine.



- Map safety tags with user-defined data type (UDT).
 - Create Standard and Safety tags of the same Data Type. A best practice is to map a UDT of information. In this example, only one bit is used.

Scope:	GLX01	Show:	All Tags	safe
Name	Alias For	Base Tag	Data Type	Class
CE_HMI_to_Controller_Safety			CE_HMI_to_Controller_Safety	Standard
CE_Active_EM_Safety			DINT	Safety
Standard_to_Safety			UDT_Mapped_Tags	Standard
Standard_to_Safety.Bool			DINT[10]	Standard
Standard_to_Safety.DINT			DINT[20]	Standard
Standard_to_Safety_Sfty			UDT_Mapped_Tags	Safety
Standard_to_Safety_Sfty.Bool			DINT[10]	Safety
Standard_to_Safety_Sfty.DINT			DINT[20]	Safety

- In the Logic>Safety Tag Mapping dialog box, enter the tag name, click Close.

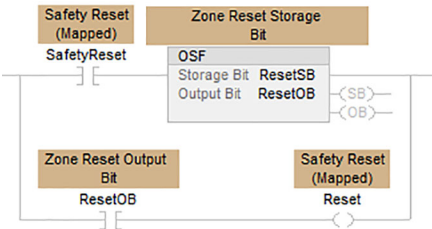
3. The safety reset signal for safety logic is mapped from the standard logic with the safety tag-mapping tool. In this example, this SafetyReset bit is in standard logic. Navigate to EM01_CE_Module01>CM04_Control_IO subroutine, Rung 0.



Falling Edge Reset

ISO 13849-1 stipulates that instruction reset functions must occur on falling edge signals. To comply with this requirement, a One Shot Falling (OSF) instruction is used on the reset rung. Then, the OSF instruction Output Bit tag is used as the reset bit for the STO output rung.

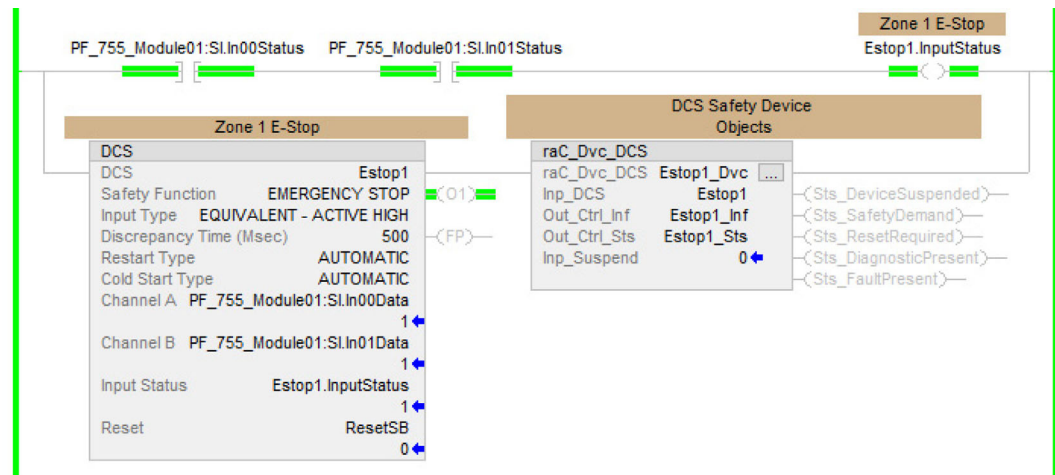
Risk Assessment and Safety Standards shall determine the location and specific requirements for safety reset.



Input (E-stop)

Table 1 - Rung 0 - E-stop Input Evaluation

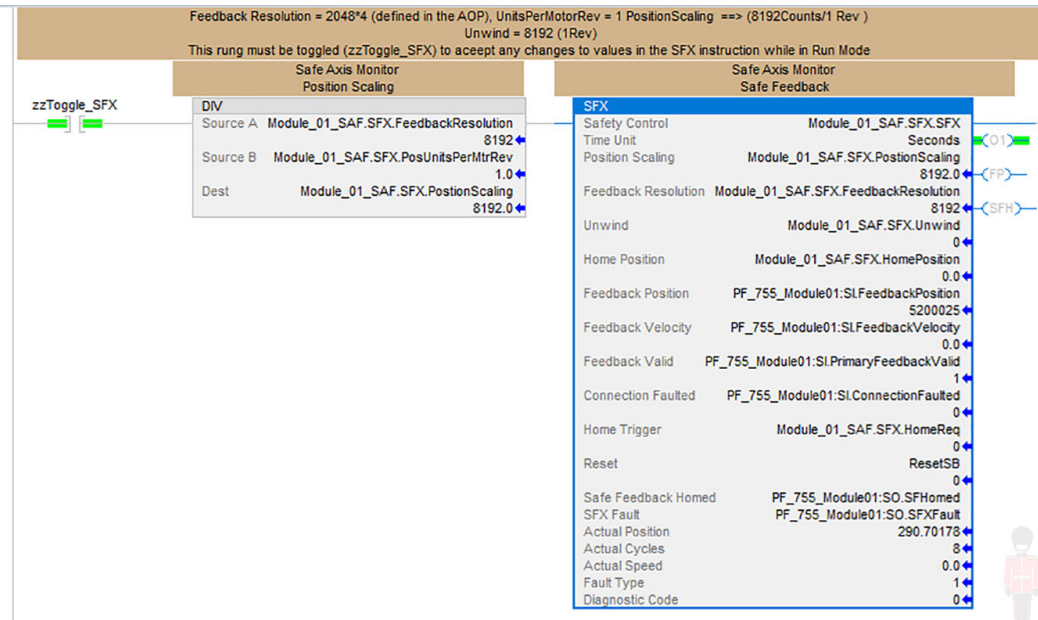
Tag	Description
InputStatus	Combines the input status of the PowerFlex inputs before the Dual Channel Input Stop tag.
DCS	Monitors the two channels for the E-stop input and the I/O status.
raC_Dvc_DCS	Connects the DCS tag to the HMI Faceplate (for example, FactoryTalk® View Machine Edition (ME), FactoryTalk View Site Edition (SE), PanelView™ 5000). The raC_Dvc_DCS Add-On Instruction tag is optional. To find this information, go to the Rockwell Automation Product Compatibility Download Center website, and search for Safety Device Library.



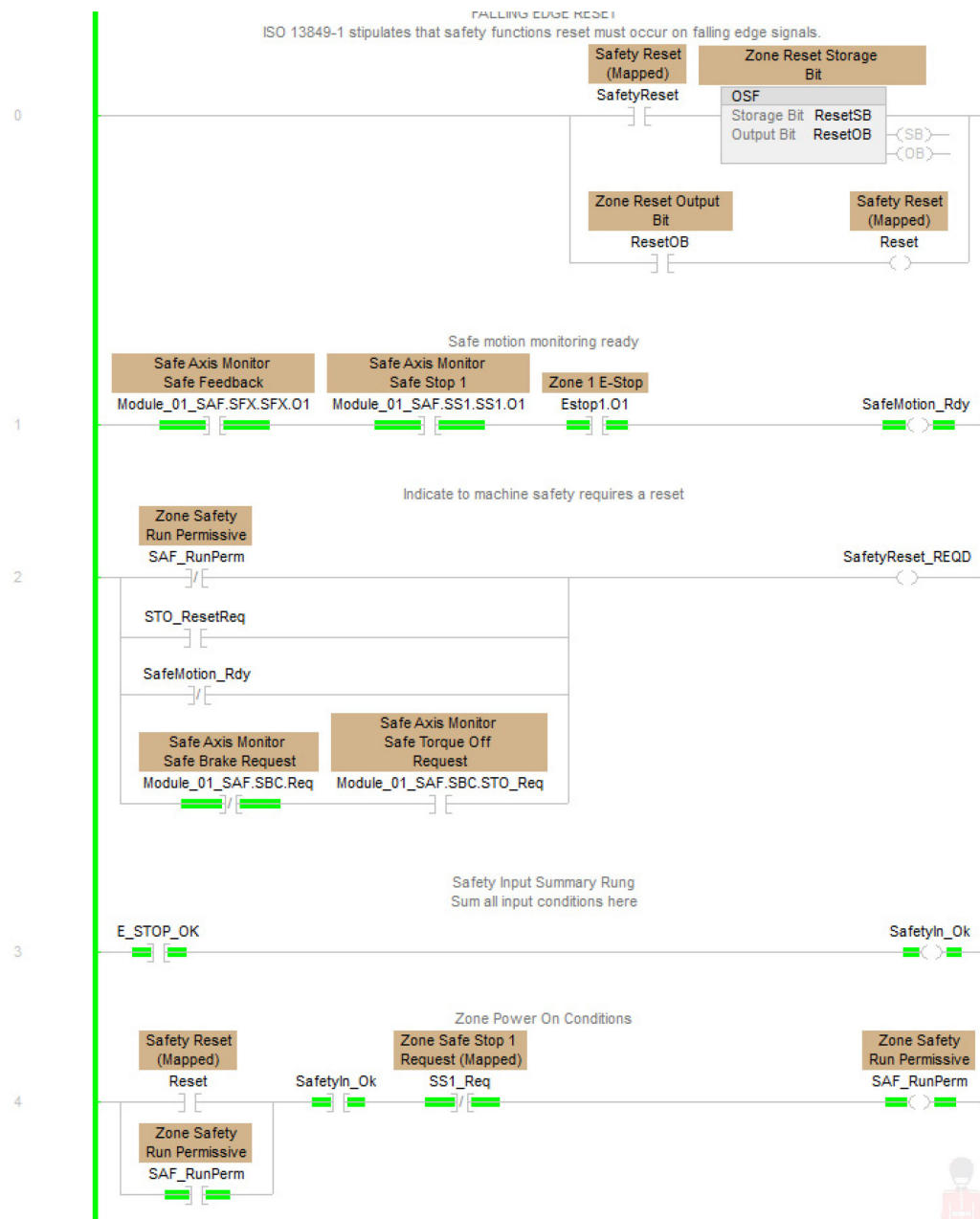
Input (SFX)

The Input SFX routine brings safe speed and position feedback into the safety task.

Tag	Description
zzToggle_SFX	Changes to the Safe Feedback Interface (SFX) configuration parameters initialize when SFX is first put on scan. Toggling the SFX rung during commissioning accepts new SFX values.
DIV	Shows feedback resolution, which comes directly from the safety module feedback properties. PositionUnitsPerMtrRev: Application scaling factor. In this example, the value is set to 1 PosUnit per motor revolution. Position Scaling: Counts/rev
SFX	Monitors safety position, velocity, and feedback status. Scales position and velocity to proper units (SFX.ActualPosition, SFX.ActualSpeed). Controls safe position homing.

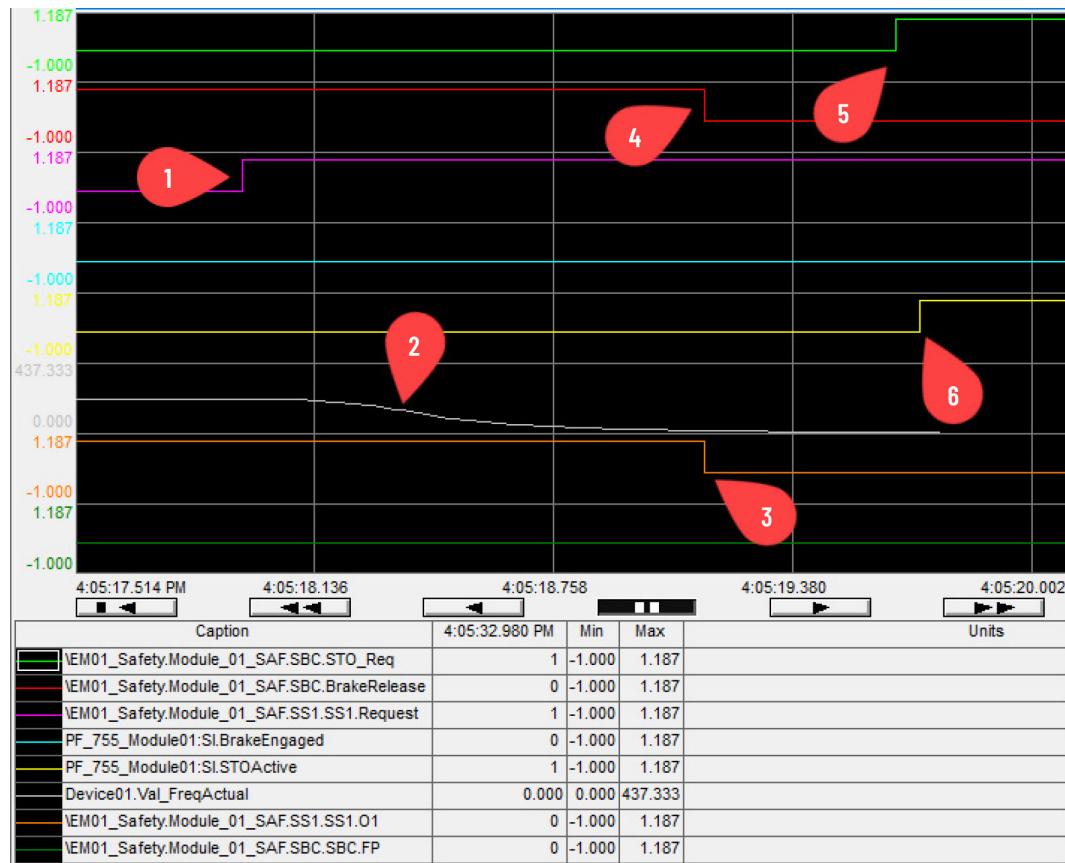


Rung	Description
0	Falling Edge Reset. This rung contains the OSF that correctly executes the Safety Reset request.
1	Evaluates the SafeMotion_Rdy condition. When the SFX, SS1 and E-stop inputs are not faulted, the SafeMotion_Rdy condition is ON. SafeMotion sums the drive safety conditions that include actual motion.
2	Evaluates different input safety conditions to activate SafetyReset_REQD condition. When this condition is active, a safety reset is required.
3	Evaluates the E-stop input condition. In this example, only the E-stop input is used, but your application can use other inputs that can be summed in this rung.
4	Run condition used for this safety zone. Summing these conditions indicates an Available or ready for Power On for this zone. In this example, one zone is used, your application can use multiple zones if necessary.



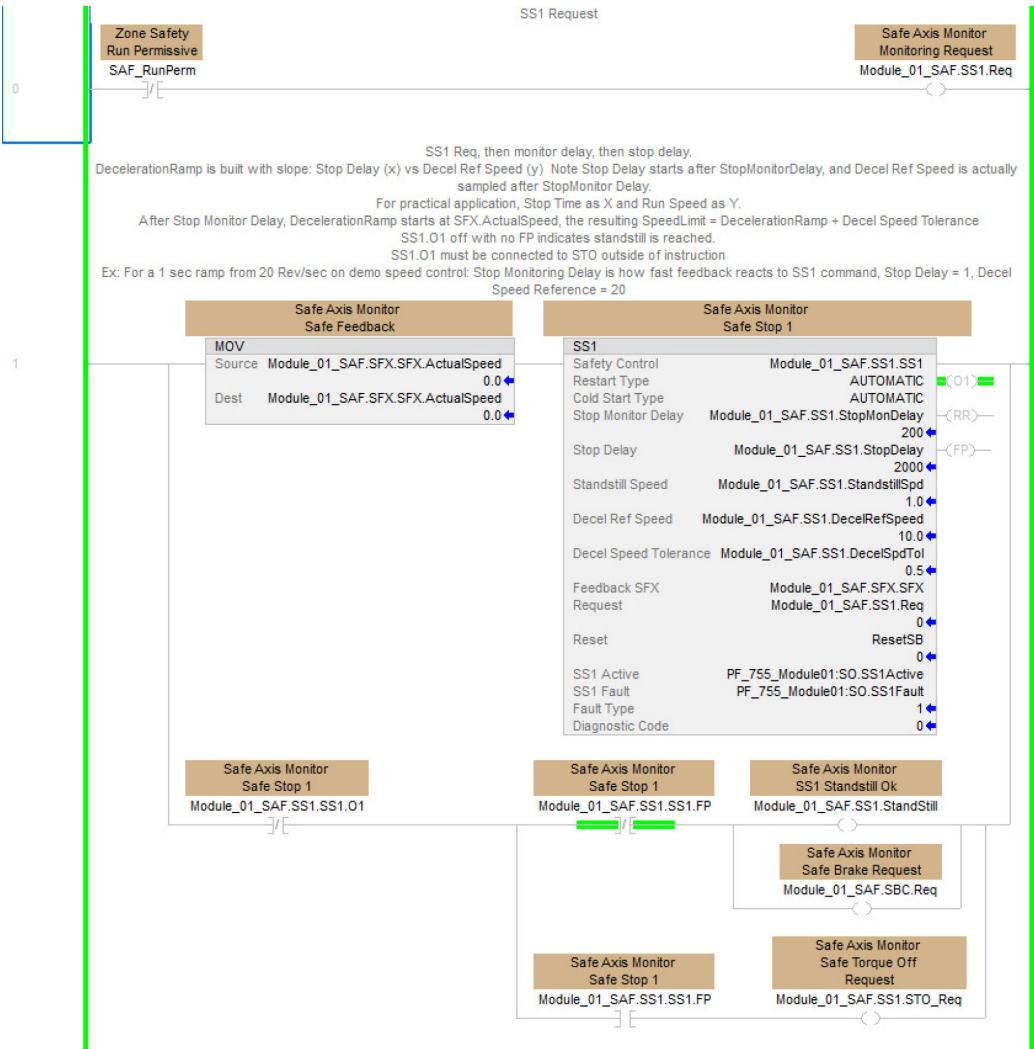
Output (SS1)

This subroutine controls when the SS1 function is used. This trend capture shows a normal (not faulted) SS1 operation.



1. SS1.Request is made from the permissive in Rung 0.
2. The motor deceleration is initiated from the standard logic (EP_Terminate subroutine) when the SS1Active pass-through tag is active.
3. Standstill speed is reached and the SS1.O1 is OFF.
4. The SBC.BrakeRelease is OFF because of the safety logic. When the SBC sequence is completed without fault, the Safety Braking Device is engaged.
5. The STO_Req (STO Request) is initiated from the SBC logic.
6. The STOActive is ON and the motor cannot produce torque.

Rung	Description
0	Evaluates the conditions that are used to initiate the SS1 function. In this case, the SAF_RunPerm is controlled from the Logic subroutine. In this example, when the E-stop input is pressed.
1	Executes and monitors the SS1 function. In this example, while the SS1 is not faulted, when the SS1.Req is ON, the SS1 configuration is used to begin the monitored deceleration. When the SS1.01 bit is OFF, the standstill condition is active. Next, the SBC request is made. When a Fault is present with the SS1 function, the STO request is ON.

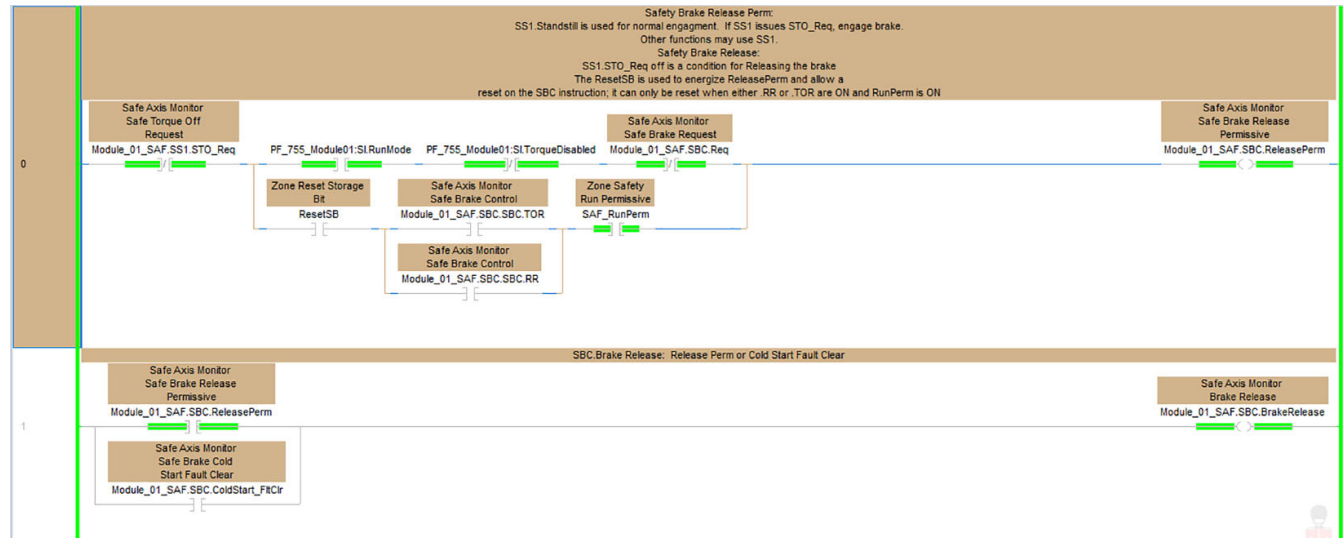


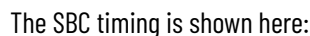
Output (SBC)

In normal operation, the SBC instruction is always monitored so that it actively controls the safety-braking device. When the safety-braking device is released, the motor can perform its motion. If any unwanted application conditions occur, the SBC engages the safety-braking device. Next, the SBC logic requests an STO function and once the STO function is active, the motor cannot produce torque and cannot generate motion.

The safety brake device engagement is application dependent and can be modified to meet your needs.

Rung	Description
0	- Safety Brake Release Perm: SS1.Standstill is used for normal engagement. If the SS1 issues STO_Req then engage the brake. Other functions may use SS1. - Safety Brake Release: SS1.STO_Req OFF is a condition for releasing the brake. The ResetSB is used to energize SBC.ReleasePerm and allow a reset on the SBC instruction; it can only be reset when either SBC.RR or SBC.TOR are ON and SAF_RunPerm is ON.
1	Evaluates the SBC.BrakeRelease bit; which is used with the SBC instruction. This rung includes the ColdStart_FltClr condition.





Our example uses a negative STO to SBC Delay. It is typical that with this negative delay, the motor is brought to a controlled stop first, then actively held at a standstill. With the motor held at standstill, SBC is activated, which applies the safety-braking device first, then after the STO to SBC delay, torque is removed.

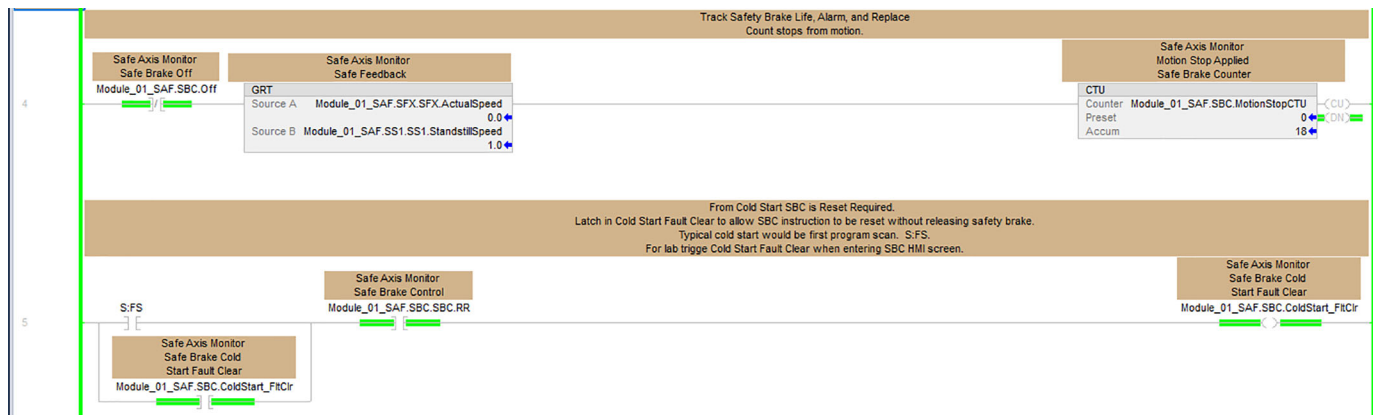
You can use positive STO to SBC Delay. It is typical that with the positive delay, a Cat 0 (uncontrolled) stop is used. With this type of stop, torque is removed from the motor first, and then, after the STO to SBC delay, the safety brake device is applied. In this case, the motor coasts before the safety-braking device is applied. More information on the SBC configuration and timing can be found in the GuardLogix Instruction Set.

When the SBC Brake outputs (B01+B02) are ON, PF_755_Module01:S0.Out00Output is ON. The K1 safety relay coil energizes, which physically releases the safety-braking device.

If the SBC instruction is faulted (SBC.FP=ON) or a typical SBC Torque Off Request occurs, an STO request is made to remove torque-producing capability from the motor.

The SBC instruction timing controls the Brake Engaged bit of the SBC instruction.

Brake Engaged Bit	Description
ON	The motor is not free for rotation and is held by the safety-braking device.
OFF	The motor can move freely and is not being held by the safety-braking device.



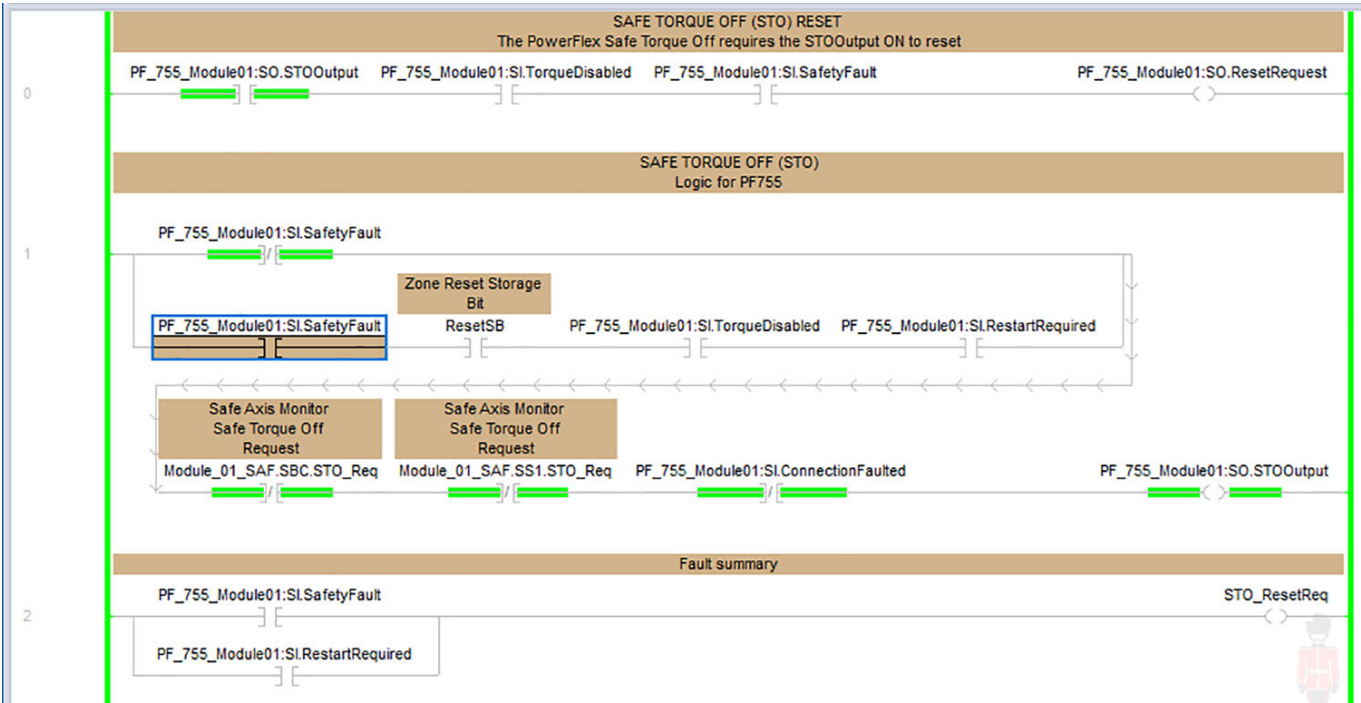
In Rung 4, the occurrence of the safety brake engaging when a non-standstill speed condition occurs is logged. This action is considered as using the safety-braking device as a stopping brake (to stop the motor rotation) and a holding brake. This action is also evaluated because a limited number of stopping-brake actuations are allowed before service and/or replacement of the safety-braking device is required. This logic counter value must be used in your application logic to verify that you do not exceed the maximum stopping-brake cycles.

Rung 5 contains the first scan and ColdStart logic to clear faults with the SBC instruction when restarting the machine, or the controller.

Output (STO)

STO (Safe Torque Off) is the safety-rated programmable function where you can allow or remove torque producing capability in the motor. When the STO function is active, the motor cannot produce torque or motion. The drive cannot be enabled. The logic that is shown here is similar to other safety logic used with different Rockwell Automation power and motion devices.

Rung	Description
0	Evaluates the conditions that are used to initiate the ResetRequest. The STOOutput bit must be on to reset the Safe Torque Off condition.
1	Evaluates different input conditions that remove the STOOutput.
2	Used to provide a Fault Summary for this zone and if a reset is required for the STO.



Programming Standard

Enable and Produce Motion on the Motor

Once the machine is started, the _App_Execute subroutine is used to Activate (Enable) and Run the motor at a constant speed. Your application logic might be different.

Decelerate and Disable the Motor

Deceleration and disabling of the motor in this example is in the _EP_Terminate subroutine. In this subroutine, the example looks for the SS1 to be active using the pass-through tags. When the SS1 is active, an AbortRequest is performed, which includes logic that decelerates and deactivates the motor. Your application logic might require a different stopping action.



Functional Safety Data Required for Determining the Performance Level of Electromechanical Devices

Because the Bulletin 800T E-stop push button is an electromechanical device, the functional safety data that is required for the Performance Level calculation includes the following:

- Mean Time to Failure, dangerous (MTTFd) = 2500
- Diagnostic Coverage (DCavg) = 99% (high)
- Common Cause Failure (CCF) = 65 (fulfilled)

The functional safety evaluations of the electromechanical devices include the following:

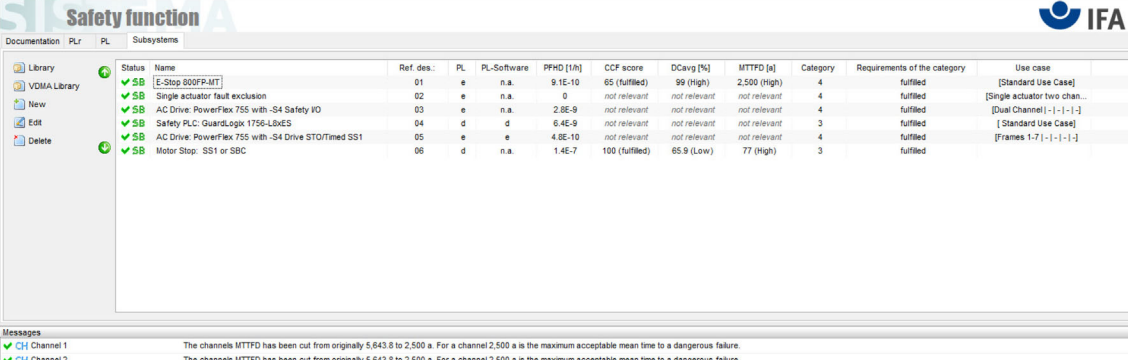
- How frequently they are operated
- Whether they are effectively monitored for faults
- Whether they are properly specified and installed

SISTEMA calculates the MTTFd by using the B10d data that is provided for the contactors along with the estimated frequency of use, entered during the creation of the SISTEMA project.

The DCavg (99%) for the contactors is selected from the Output Device table of ISO 13849-1 Annex E, Direct Monitoring.

The DCavg (99%) for the E-stop is selected from the Input Device table of ISO 13849-1 Annex E, Cross Monitoring.

The CCF value is generated by using the scoring process that is outlined in Annex F of ISO 13849-1. The complete CCF scoring process must be performed when actually implementing an application. A minimum score of 65 must be achieved.



The screenshot shows the 'Safety function A' interface with a table of safety functions. The table includes columns for Status, Name, Ref. des., PL, PL-Software, PPHD [1/h], CCF score, DCavg [%], MTTFD [a], Category, Requirements of the category, and Use case. The table lists five safety functions, all of which are fulfilled.

Status	Name	Ref. des.	PL	PL-Software	PPHD [1/h]	CCF score	DCavg [%]	MTTFD [a]	Category	Requirements of the category	Use case
✓ SB	E-Stop 800FP-MT	01	e	n.a.	9.1E-10	65 (fulfilled)	99 (High)	2,500 (High)	4	fulfilled	[Standard Use Case]
✓ SB	Single actuator fault exclusion	02	e	n.a.	0	not relevant	not relevant	not relevant	4	fulfilled	[Single actuator two chan...]
✓ SB	AC Drive: PowerFlex 755 with -S4 Safety I/O	03	e	n.a.	2.8E-9	not relevant	not relevant	not relevant	4	fulfilled	[Dual Channel] - [-] [-] [-]
✓ SB	Safety PLC: GuardLogix 1756-L84S	04	d	d	6.4E-9	not relevant	not relevant	not relevant	3	fulfilled	[Standard Use Case]
✓ SB	AC Drive: PowerFlex 755 with -S4 Drive STO/Timed SS1	05	e	e	4.8E-10	not relevant	not relevant	not relevant	4	fulfilled	[Frames 1-7] - [-] [-] [-]
✓ SB	Motor Stop: SS1 or SBC	06	d	n.a.	1.4E-7	100 (fulfilled)	65.9 (Low)	77 (High)	3	fulfilled	

Messages:

- ✓ CH Channel 1 The channels MTTFD has been cut from originally 5,643.6 to 2,500 a. For a channel 2,500 a is the maximum acceptable mean time to a dangerous failure.
- ✓ CH Channel 2 The channels MTTFD has been cut from originally 5,643.6 to 2,500 a. For a channel 2,500 a is the maximum acceptable mean time to a dangerous failure.
- ✓ CH Channel 1 The channels MTTFD has been cut from originally 268.7 to 100 a. For a channel 100 a is the maximum acceptable mean time to a dangerous failure.

When an application includes two-channel, single-actuator mechanical safeguarding devices such as interlocks or mechanical emergency stop devices such as E-stops or cable pull switches, fault exclusion for single-actuator failure must be accounted for and, when required, applied when determining the Performance Level.

Exclusion of the possible fault of the single actuator failing to switch the two channels properly is not allowed. Therefore, single types of electromechanical devices are limited to a maximum Performance Level of d. The Performance Level required (PLr) in safety function application techniques is PLd. Redundancy of safeguarding switches is required to achieve Performance Level e.

If the maximum number of operations of an electromechanical E-stop device is in accordance with IEC 60947-5-5, regarding the mechanical aspects of the device, exclusion of the possible fault of the single actuator of that device failing to switch the two channels properly is allowed per EN ISO 13849-2, Annex D, Table D8. Therefore, single types of devices, properly applied, are not limited and can achieve Performance Level e.

The emergency stop function is a complementary protective measure that is intended to be used with other safeguarding measures and protective devices to reduce risk sufficiently. The design of the emergency stop functions must not impair the effectiveness of other safety functions or protective devices in the system. The actual number of operations (NOP) is used for the purposes of the MTTFd calculation in this document.

Verification and Validation Plan

Verification and validation play important roles in the avoidance of faults throughout the safety system design and development process. ISO 13849-2 sets the requirements for verification and validation. The standard calls for a documented plan to confirm that all safety functional requirements have been met.

Verification is an analysis of the resulting safety control system. The Performance Level (PL) of the safety control system is calculated to confirm that the system meets the required Performance Level (PLr) specified. The SISTEMA software is typically used to perform the calculations and assist with satisfying the requirements of ISO 13849-1.

Validation is a functional test of the safety control system to demonstrate that the system meets the specified requirements of the safety function. The safety control system is tested to confirm that all safety-related outputs respond appropriately to their corresponding safety-related inputs. The functional test includes normal operating conditions and potential fault injection of failure modes. A checklist is typically used to document the validation of the safety control system.

Before validating the GuardLogix Safety System, confirm that the safety system and safety application program have been designed in accordance with the controller safety reference manuals that are listed in [Additional Resources](#) and the GuardLogix Safety Application Instruction Set Reference Manual, publication [1756-RM095](#).

For a validation checklist, see the attached spreadsheet. For instructions on how to access the attachments, see [Use Sample Project Files on page 5](#).

Additional Resources

These documents contain additional information about related products from Rockwell Automation.

Resource	Description
GuardLogix 5580 and Compact GuardLogix 5380 Controller Systems Safety Reference Manual, publication 1756-RM012	Describes the GuardLogix 5580 and Compact GuardLogix 5380 controller system. Provides instructions on how to develop, operate, or maintain a controller-based safety system that uses the Studio 5000 Logix Designer application.
ControlLogix 5580 and GuardLogix 5580 Controllers User Manual, publication 1756-UM543	Provides information on how to install, configure, and program the GuardLogix 5580 controllers in the Logix Designer application.
CompactLogix 5380 and Compact GuardLogix 5380 Controllers User Manual, publication 5069-UM001	Provides information on how to install, configure, and program the Compact GuardLogix 5380 controllers in the Logix Designer application.
GuardLogix Safety Application Instruction Set Reference Manual, publication 1756-RM095	Describes the Rockwell Automation GuardLogix Safety Application Instruction Set. Provides instructions on how to design, program, or troubleshoot safety applications that use GuardLogix controllers.
PowerFlex 750-Series I/O, Feedback, and Power Option Modules Installation Instructions, publication 750-IN111	Contains wiring, configuration, and installation instructions for the PowerFlex 755 drive option cards.
PowerFlex 755/755T Integrated Safety Functions Option Module User Manual, publication 750-UM005	Contains wiring, configuration, and installation instructions for the PowerFlex 755 drive safety option card and add-on profile configurable functions.
Relay and Timer Specifications Technical Data, publication 700-TD552	Provides technical documentation about the Bulletin 700S-CF safety relays.
Rockwell Automation Functional Safety Data Sheet, publication SAFETY-SR001	Provides functional safety data for Rockwell Automation® products.
Industrial Automation Wiring and Grounding Guidelines, publication 1770-4.1	Provides general guidelines for installing a Rockwell Automation industrial system.
Product Certifications website, rok.auto/certifications	Provides declarations of conformity, certificates, and other certification details.
Safety Automation Builder and SISTEMA Library website, rok.auto/sistema	Download Safety Automation Builder® to help simplify machine safety design and validation, and reduce time and costs. Integration with our risk assessment software provides you with consistent, reliable, and documented management of the Functional Safety Lifecycle. The SISTEMA tool, also available for download from the Safety Automation Builder page, automates calculation of the attained Performance Level from the safety-related parts of a machine's control system to (EN) ISO 13849-1.

You can view or download publications at [rok.auto/literature](#).

Rockwell Automation Support

Use these resources to access support information.

Technical Support Center	Find help with how-to videos, FAQs, chat, user forums, Knowledgebase, and product notification updates.	rok.auto/support
Local Technical Support Phone Numbers	Locate the telephone number for your country.	rok.auto/phonesupport
Technical Documentation Center	Quickly access and download technical specifications, installation instructions, and user manuals.	rok.auto/techdocs
Literature Library	Find installation instructions, manuals, brochures, and technical data publications.	rok.auto/literature
Product Compatibility and Download Center (PCDC)	Download firmware, associated files (such as AOP, EDS, and DTM), and access product release notes.	rok.auto/pcdc

Documentation Feedback

Your comments help us serve your documentation needs better. If you have any suggestions on how to improve our content, complete the form at rok.auto/docfeedback.

Waste Electrical and Electronic Equipment (WEEE)



At the end of life, this equipment should be collected separately from any unsorted municipal waste.

Rockwell Automation maintains current product environmental compliance information on its website at rok.auto/pec.

Safety Function Capabilities

Visit rok.auto/safety for more information on our Safety System Development Tools, including [Safety Functions](#).

Allen-Bradley, CompactLogix, ControlLogix, expanding human possibility, FactoryTalk, Guard I/O, GuardLogix, PanelView, PowerFlex, Rockwell Automation, Safety Automation Builder, SensaGuard, and Studio 5000 Logix Designer are trademarks of Rockwell Automation, Inc.

CIP is trademarks of ODVA, Inc.

Trademarks not belonging to Rockwell Automation are property of their respective companies.

Rockwell Otomasyon Ticaret A.Ş. Kar Plaza İş Merkezi E Blok Kat:6 34752, İçerenkÖy, İstanbul, Tel: +90 (216) 5698400 EEE Yönetmeliğine Uygundur

Connect with us.    

rockwellautomation.com — expanding **human possibility**®

AMERICAS: Rockwell Automation, 1201 South Second Street, Milwaukee, WI 53204-2496 USA, Tel: (1) 414.382.2000

EUROPE/MIDDLE EAST/AFRICA: Rockwell Automation NV, Pegasus Park, De Kleetlaan 12a, 1831 Diegem, Belgium, Tel: (32) 2663 0600

ASIA PACIFIC: Rockwell Automation SEA Pte Ltd, 2 Corporation Road, #04-05, Main Lobby, Corporation Place, Singapore 618494, Tel: (65) 6510 6608

UNITED KINGDOM: Rockwell Automation Ltd., Pitfield, Kiln Farm, Milton Keynes, MK11 3DR, United Kingdom, Tel: (44)(1908) 838-800